AURORA-Mono (One-Piece)

Rover Wheel — Build Specification

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Abstract

AURORA-Mono is a one-piece composite rover wheel engineered for the MicroChariot interface envelope. The design combines a carbon-nanotube-reinforced PEKK structural cage with a silicon-carbide-filled PEKK wear tread, joined by a co-molded mechanical and chemical bond. This document specifies the envelope, materials, internal lattice geometry, hub interface, tread system, manufacturing sequence, and as-built mass targets required to fabricate a flight-representative unit.

0. Envelope & Interfaces (MicroChariot Compliant)

- Overall OD at lug tips: **18.000 in** (457.20 mm)
- Overall width: **8.000 in** (203.20 mm)
- Rim OD (outer structural skin, pre-tread): **16.800 in** (426.72 mm)
- Hub pilot: **Ø 3.185 in** (80.899 mm)
- Bolt circle: **Ø 4.000 in** (101.60 mm), 8× #10-32 UNF THRU, equally spaced (45°)
- Jack bolts: 2× #10-32 UNF THRU on the same Ø 4.000 in circle, 180° apart
- Pin holes: 2× Ø 0.2502 ±0.001 in (6.356 ±0.025 mm) on Ø 4.000 in circle, 180° apart
- Hub adapter plate thickness (composite): 6.0 mm with 12.0 mm diameter pad bosses at holes
- Keep-out zone: honors 4.000 in (W) × 1.750 in (H) box from NASA drawing; no intrusion

Tolerances (unless noted)

- Hub pattern features: ±0.05 mm (±0.002 in)
- Critical radii (OD, width): ±0.25 mm (±0.010 in)
- All others: ±0.5 mm (±0.020 in)

1. Materials

- Structural skins & ribs: **Carbon Nanotube-reinforced, Carbon Fiber-reinforced Polyetherketoneketone (PEKK-CNT/CF)**, ρ ≈ 1.45 g/cc
- Tread lugs (wear layer): **Silicon Carbide-filled Polyetherketoneketone (SiC-PEKK)**, ρ ≈ 1.65 g/cc
- Hub hole sleeves (optional): Ti-6Al-4V, wall 0.5 mm (press-fit) for metal interfaces
- Antistatic top-coat (non-lug areas): graphene-based, 0.05 mm (vacuum-compatible)

2. Rim Wall “Sandwich” Cross-Section

All dimensions radial from wheel center.

- Outer structural skin (bonding surface under tread): **1.20 mm** thick, continuous and sealed
- Core height between skins: **7.00 mm** nominal
- Inner structural skin (toward hub): **1.20 mm** thick, continuous
- Wall total (skin + core + skin): **9.40 mm**
- Inner surface local relief under lugs (flex zone): reduce inner-skin-side wall to 0.70–0.90 mm within a 28 × 20 mm oval directly beneath each lug, machined/pocketed from inside

Helical Ribs (Between Skins)

- Pattern: alternating ±35° helix families (X-brace lattice)
- Rib web thickness: **1.8 mm** at mid-span, **2.2 mm** at skin tie-in
- Rib pitch (circumferential spacing of nodes): 20.0 mm between successive ties on each skin
- Rib bay (axial spacing across width): 12.0 mm, yielding ≈17 bays across the 203.20 mm width
- Count around circumference: **48 ribs total** (24 at +35°, 24 at –35°), evenly distributed
- Local neck-down beneath lug footprints: –12% rib thickness over a 20.0 mm arc length (adds micro-compliance)

Inboard Debris Ports (NOT on tread face)

- Ø 10.0 mm, one every 35° (≈10 ports per wheel)
- Lip-baffled with a 2.0 mm deep overhang; no line-of-sight to the outer skin

3. Hub “Spokes” (Torque Web, Inside Cavity)

Thin composite webs run from the hub adapter to the inner rim skin and are not visible in the tread wall.

- Count: **6 web-spokes**, equally spaced
- Web thickness: 5.0 mm at hub boss tapering to 3.0 mm at rim inner skin; aero-tri profile with 5.0 mm fillets both ends
- Lightening pockets: triangular cutouts leaving ≥30.0 mm minimum shear path; web area open ratio ≈45%
- Purpose: torque transfer and stiffness with minimal mass

4. Tread System

Integral tread, molded onto the sealed outer skin. Base (carcass) thickness to reach the 18.000 in OD with 9.0 mm lugs is t_{base} = **6.24 mm** (from rim outer skin up to the lug root). The construction places a thin outer shell (~1.0 mm) over the lattice peaks; the remaining ~5.2 mm is mostly rib crests and void, yielding a 10–12% solid equivalent.

Lug Geometry (Two Staggered Rows, Dirt-Bike Style)

- Height: 9.0 mm nominal; alternating 8.0 mm / 10.0 mm every other lug within a row
- Plan shape: chevron blocks; rake angle 60° leading face, trailing chamfer 22°
- Block length (circumferential): 45.0 mm long / 35.0 mm short, alternating (±5.0 mm)
- Block width (axial): 14.0 mm; row centers at ±30.0 mm from mid-plane (offset across the 8.000 in width)
- Row phase shift (helical wrap): ½-pitch (≈20.0 mm) between rows
- Shoulder side-biters: every 3rd lug gets a 3.0 mm tall, 4.0 mm long buttress at the outer shoulder
- Void ratio (contact patch): 75–80% (coverage 19–20%)
- Root fillets: 1.2 mm on all interior edges to avoid chipping
- External corner radii: 0.6–0.8 mm (keeps edges sharp but durable)

Anti-Peel Features (With Sealed Skin)

- Micro keys: 3 concentric 0.6 mm deep × 1.8 mm wide shallow grooves in the outer skin surface under each lug; these fill with SiC-PEKK during molding (mechanical key)
- Chemical bond: same PEKK family; over-mold at 355–365 °C and 0.6–0.8 MPa dwell ≥20 min

5. Manufacturing Notes (Sequence)

1. Print lattice core and hub web in PEKK-CNT/CF using high-temperature additive manufacturing.
2. Lay inner/outer skins (PEKK-CNT/CF tape) and co-consolidate in an autoclave (≈360 °C, 0.7 MPa, 2–3 hours).
3. Machine the hub pattern, debris ports, and inner-surface relief pockets under the lug locations.
4. Compression-mold the SiC-PEKK tread onto the sealed outer skin using the chevron cavity tool (engaging the anti-peel keys).
5. Apply the antistatic top-coat, avoiding lug faces.
6. Final QC: dimensional validation, mass validation (target 2.21–2.30 kg), and non-destructive inspection (ultrasound / CT).

6. Critical Dimensions Summary

Feature	Dimension
OD at lug tips	18.000 in (457.20 mm)
Width	8.000 in (203.20 mm)
Rim OD (pre-tread)	16.800 in (426.72 mm)
Outer / inner skin	1.20 mm each
Core height	7.00 mm
Helical rib web	1.8 mm (mid-span), 2.2 mm (tie-in)
Helix angle	±35°

Feature	Dimension
Ribs (count)	48 total (24 positive / 24 negative)
Hub web spokes	6 spokes, 3.0–5.0 mm thick
Tread base (carcass)	6.24 mm (10–12% solid equivalent)
Lug height	9.0 mm nominal (8.0 / 10.0 mm alternating)
Lug width	14.0 mm
Lug length (pitch)	35.0 / 45.0 mm alternating
Row offset	±30.0 mm from mid-plane
Void ratio	75–80% (coverage 19–20%)
Anti-peel keys	3 grooves, 0.6 mm × 1.8 mm
Debris ports (inboard)	Ø 10.0 mm, every 35°, baffled

Table 1. Critical dimensions summary.

7. Mass Target (As-Built)

- Nominal specification: ≈ **2.21–2.30 kg** (4.87–5.07 lb)
- Mass optimization margin: reduce total surface coverage down to 19%, or implement hollow-web configurations within the lugs (saving 80–120 g) while maintaining the identical external footprint